STUDENT LEARNING PERFORMANCE AND ACADEMIC RISK ANALYSIS

Formal Analytical Report

EduRankAI – Data Science & Analytics, Domain 02

Dataset: Student_Performance.csv

Purpose: Examine associations between student learning behaviour, background characteristics and academic performance, and describe a transparent screening approach for identifying students who may need additional academic support.

Executive Summary

The supplied CSV contains 25,000 rows and 16 columns. It contained 10,000 exact duplicate rows. After exact duplicate removal, 15,000 records remained, with no duplicate student IDs and no missing values.

Study hours had a strong positive association with overall score (Pearson r = 0.906, 95% CI [0.903, 0.909], p < 0.001). Attendance had a weaker positive association (r = 0.288, p < 0.001). The Kruskal–Wallis test did not identify statistically significant score-distribution differences across the six study methods (H = 10.205, p = 0.070).

For descriptive screening, grades E and F were grouped as “At Risk”. This provisional rule classified 5,174 students (34.49%) in the cleaned dataset. It is not an institutionally validated risk policy.

These are observational associations. They do not establish that increasing study hours or attendance causes academic scores to improve.

1. Problem and Decision Context

Educational institutions need to understand patterns in student performance and identify learners who may benefit from additional support. This analysis examines how study hours, attendance and selected background characteristics vary with academic scores. Its intended use is to inform further investigation and supportive planning, not to automate high-stakes decisions about individual students.

2. Research Question and Objectives

Primary question: Which student learning and background factors are associated with academic performance, and how can students at potential academic risk be identified using their learning behavior and academic scores?

Objectives:

- Assess dataset quality.
- Describe academic score distributions.
- Examine associations between study hours, attendance and overall score.
- Compare score distributions across study methods.
- Describe the size and characteristics of a provisional E/F risk group.

3. Hypotheses

- H1: Study hours are linearly associated with overall score.
- H2: Attendance percentage is linearly associated with overall score.
- H3: Overall-score distributions differ across study-method groups.

The null hypotheses state no association for H1/H2 and no group-distribution difference for H3. The significance level was $\alpha = 0.05$.

4. Data Sources and Data Description

The source file for this analysis is the user-supplied Student_Performance.csv. It contains 16 fields: student_id, age, gender, school_type, parent_education, study_hours, attendance_percentage, internet_access, travel_time, extra_activities, study_method, math_score, science_score, english_score, overall_score and final_grade.

The file does not provide a source URL, collection dates, sampling frame, data-collection method, grading rubric or formal data dictionary; provenance and representativeness therefore cannot be independently verified.

After cleaning, the dataset contains 15,000 records. Age ranges from 14 to 19 years; study hours range from 0.5 to 8.0; attendance ranges from 50.0% to 100.0%. Overall score ranges from 14.5 to 100.0, with mean 64.02 and median 64.30.

5. Data Quality Assessment and Preparation

The raw file had 25,000 rows. Exact duplicate checking found 10,000 duplicate rows. These exact duplicates were removed using row-level duplicate detection, leaving 15,000 records and 0 repeated student IDs. The cleaned data contained 0 missing values.

Observed numeric ranges were checked: age 14–19, attendance 50–100%, and subject/overall scores 0–100.

Cleaning decision: exact duplicate rows were removed because all fields in the repeated rows matched. No imputation was required because no missing values remained. The original values and category labels were retained.

6. Methodology

Descriptive analysis: counts, percentages, means and medians summarized the data. Pearson correlation measured linear associations between study hours/attendance and overall score; Spearman rank correlation provided a robustness comparison. The Kruskal–Wallis test compared overall-score distributions across the six independent study-method groups. For risk screening, final grades E and F were grouped into a provisional “At Risk” category.

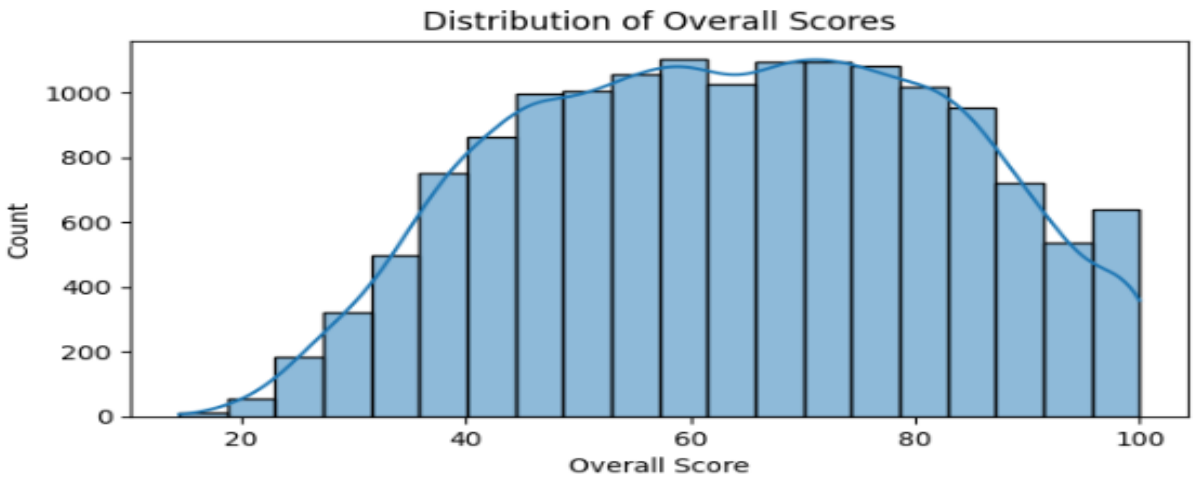
The significance threshold was $\alpha = 0.05$. P-values are interpreted alongside effect magnitude and practical meaning. A small p-value in a large dataset does not by itself imply a practically important relationship.

7. Results and Findings

7.1 Overall Score Profile

The cleaned dataset mean overall score was 64.02 (SD 18.98); median was 64.30. Grade distribution: A 4.81%, B 10.92%, C 24.65%, D 25.13%, E 22.52%, F 11.97%.

Study Hours and Overall Score



Study hours showed a strong positive Pearson correlation with overall score ($r = 0.906$, $p < 0.001$; 95% CI [0.903, 0.909]). Spearman correlation was also strong ($\rho = 0.911$, $p < 0.001$). The agreement supports a strong monotonic association in this dataset, but it does not demonstrate causation.

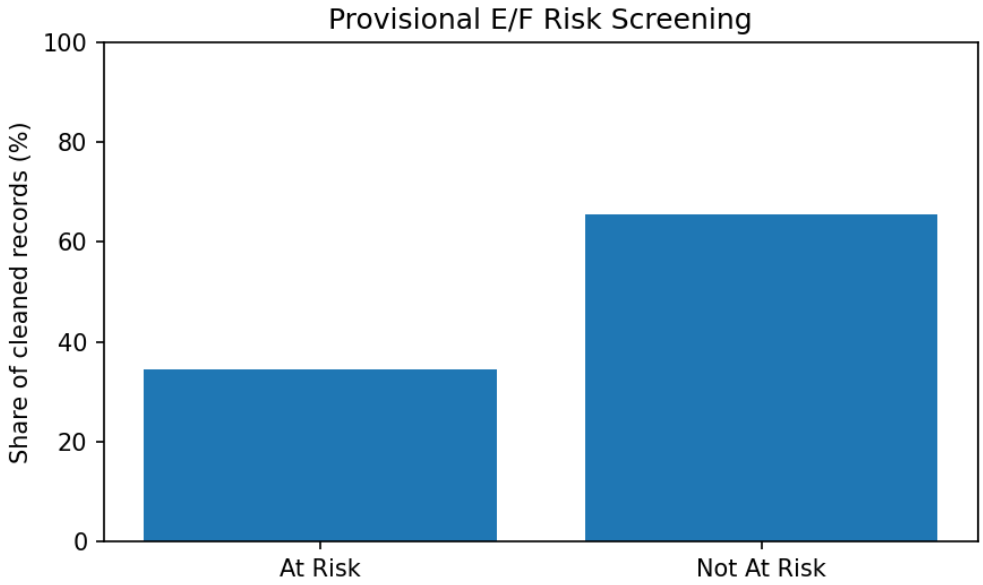
Attendance and Overall Score

Attendance had a positive but weaker association with overall score (Pearson $r = 0.288$, $p < 0.001$; Spearman $\rho = 0.273$, $p < 0.001$). Its magnitude is substantially smaller than the study-hours association and should be interpreted as modest.

Study-Method Comparison

Mean overall scores by study method: online videos: 64.68; coaching: 64.48; textbook: 64.19; notes: 63.86; mixed: 63.65; group study: 63.25. The Kruskal–Wallis test was not statistically significant at $\alpha = 0.05$ ($H = 10.205$, $p = 0.070$; epsilon-squared estimate ≈ 0.00035). The analysis therefore does not provide sufficient evidence that score distributions differ across study-method categories. Descriptive differences should not be treated as proof that one method is better.

Provisional Academic-Risk Screening



Using the provisional E/F rule, 5,174 of 15,000 students (34.49%) were classified At Risk; 9,826 (65.51%) were Not At Risk.

Mean study hours were 1.98 for At Risk and 5.46 for Not At Risk. Mean attendance was 71.18% and 77.00%, respectively. Mean overall score was 42.79 in the At Risk group and 75.19 in the Not At Risk group.

This score difference is expected to be large because the groups were defined using final grades that are related to academic performance. It is descriptive, not independent evidence of predictive performance.

Risk rates by attendance band decreased descriptively: (-0.001, 60.0]: 47.79%; (60.0, 75.0]: 38.88%; (75.0, 90.0]: 29.28%; (90.0, 100.0]: 22.24%. These unadjusted percentages do not establish a causal attendance effect.

School Type and Parent Education

Mean overall score was 64.04 for private-school records and 63.99 for public-school records, a descriptive difference of 0.04 points. Average scores by parent-education category ranged from 63.45 to 64.77. These unadjusted comparisons are not causal and do not control for other variables.

8. Validation and Robustness

First, the cleaned dataset contained 15,000 records with no repeated student IDs. Second, Pearson and Spearman correlations for study hours were similar ($r = 0.906$; $\rho = 0.911$), so the positive association was not dependent only on Pearson's linear correlation measure. Third, a sensitivity check excluded records outside the 1st–99th percentile ranges of both study hours and overall score. 14,673 records remained and the Pearson correlation was 0.902, compared with 0.906 in the full cleaned data; the association remained strong.

For attendance, Pearson $r = 0.288$ and Spearman $\rho = 0.273$, both indicating a positive but modest association. The study-method test returned $p = 0.070$, so the null hypothesis was not rejected at $\alpha = 0.05$. These checks improve transparency but do not replace independent replication on a separately collected dataset.

9. Interpretation and Recommendations

- Study hours show a strong association with overall score. Educators may investigate structured study planning and learning-support programmes, then evaluate outcomes prospectively rather than assuming they work.
- Attendance shows a weaker positive association, and the descriptive risk rate is lower in higher-attendance bands. Schools may review attendance barriers and provide appropriate follow-up while monitoring subsequent outcomes.
- The analysis did not find statistically significant score-distribution differences across study methods. Do not rank or prescribe methods based on these results; collect richer evidence and evaluate methods with a suitable design.
- The provisional E/F rule flags about one-third of records. Use this only as a starting point for human review and supportive outreach. Confirm grade definitions and risk thresholds with the institution before operational use.

10. Conclusion

After removing 10,000 exact duplicate rows, the analysis examined 15,000 student records. Study hours were strongly positively associated with overall score; attendance showed a weaker positive association. No statistically significant difference in score distributions was detected across study methods. The provisional E/F screening rule classified 34.49% of records as potentially at risk, but this rule requires institutional validation. Next steps should include verifying source provenance and variable definitions, conducting adjusted analyses and evaluating support interventions with longitudinal or prospective data.

11. Tools, Assistance and Reproducibility

Tools used: Python, pandas, NumPy, SciPy and Matplotlib. The report was prepared with AI assistance for analytical structuring and drafting. The candidate should rerun the notebook, verify all outputs and be able to explain each cleaning decision, method and conclusion before submission. No external dataset or external research source was used in this report.

Appendix A – Key Statistical Outputs

Analysis	Result	Interpretation
Study hours vs overall score	$r = 0.906$; $p < 0.001$	Strong positive association
Study hours Spearman	$\rho = 0.911$; $p < 0.001$	Similar rank association
Attendance vs overall score	$r = 0.288$; $p < 0.001$	Modest positive association
Study-method Kruskal–Wallis	$H = 10.205$; $p = 0.070$	Not significant at 0.05
Study-hours sensitivity	$r = 0.902$; $n = 14{,}673$	Association remains strong
Provisional E/F risk group	34.49%	Descriptive screening only

Appendix B – Reproducibility Note

Exact duplicate removal: df_clean = df.drop_duplicates().copy(). The provisional risk grouping used final_grade values E and F. Statistical methods: Pearson correlation, Spearman rank correlation and Kruskal–Wallis. Keep the original CSV unchanged and submit the cleaned CSV and executed analysis notebook with this report.

Appendix C – Assessment Alignment

This report covers the assessment brief's requested question and hypotheses, data description, data-quality assessment, methodology, findings, validation and robustness, interpretation, limitations, recommendations and tool/assistance declaration. The candidate should be ready to defend question selection, data choice, cleaning decisions, method choice and validation.